

Technical Design Specification

300G VERTICAL SPINNER COMBAT ROBOT | INDIA FIRST CATEGORY

1. System Architecture Overview

This document outlines the hardware architecture, component integration, and mass distribution for a 300g combat robot. The design relies on a high-discharge power system to deliver rapid kinetic energy transfer through a vertical rotating blade, utilizing repurposed multi-rotor hardware for weapon drive and high-torque micro gearmotors for locomotion.

2. Power Subsystem

The power delivery system is bottlenecked by the weapon's current draw upon spin-up and impact. To prevent voltage sag and receiver brownouts, a high-discharge lithium-polymer cell is utilized.

- **Battery Chemistry:** Lithium Polymer (LiPo)
- **Configuration:** 2S (7.4V nominal, 8.4V fully charged)
- **Discharge Rating:** 85C continuous

Engineering Note: Current Handling

The 85C rating on a ~300mAh battery allows for a continuous current draw of over 25 Amps. This is critical for instantaneous current spikes when the BLDC motor stalls against an opponent's chassis. A dedicated XT30 connector should be used to handle the current without resistive heating.

3. Weapon Subsystem: Vertical Spinner

The primary offensive capability is a vertically mounted asymmetric or symmetric blade. Vertical spinners exploit the arena floor as a reaction surface, launching opponents upward to maximize internal shock and fall damage.

Component	Specification
Weapon Motor	Outrunner BLDC motor (Drone stator, typically 1104 to 1806 size)
Motor Kv Rating	~2300Kv - 4000Kv (Optimized for 2S voltage for high tip speed)
Weapon ESC	15A - 20A Brushless ESC (BLHeli_S firmware recommended)
Blade Material	Hardened Tool Steel (e.g., AR500) or Titanium (Grade 5)

Kinematic Delivery: The BLDC outrunner provides excellent torque density. The motor can either be direct-drive (blade mounted directly to the spinning bell) or belt-driven (using a polyurethane O-ring belt to isolate the motor bearings from the destructive radial loads of weapon impacts).

4. Locomotion Subsystem

Agility and pushing power are handled by high-reduction micro metal gearmotors. Because the weapon consumes a large percentage of the weight budget, lightweight drive actuators are mandatory.

- **Actuators:** N20 Micro Metal Gearmotors (typically 2 or 4 depending on chassis layout).
- **Gear Ratio:** 30:1 to 50:1 (Targeting 400 - 600 RPM at 7.4V for optimal arena navigation speed).
- **Drive ESC:** Dual-channel brushed motor controller (e.g., Malinkent or custom dual-brushed ESC) capable of mixing differential steering.
- **Wheels:** Foam or custom cast silicone tires to absorb shock and protect N20 gearboxes.

5. Communications & Control

Signal integrity is maintained via the FlySky 2.4GHz ecosystem, providing robust interference resistance in an environment with high electromagnetic noise from brushed motor arcing and high-current BLDC switching.

Subsystem	Hardware	Notes
Transmitter (Tx)	FlySky FS-i6X	Programmed with elevon/v-tail mixing for single-stick drive control.
Receiver (Rx)		10-channel receiver utilizing AFHDS 2A protocol.

SUBSYSTEM	HARDWARE	NOTES
	FlySky FS-iA10B	
Failsafe	Rx Integrated	Must be programmed to cut weapon ESC throttle to 0% upon signal loss.

Weight Reduction Warning: FS-iA10B Receiver

The FS-iA10B is relatively large (15g) and bulky for a 300g bot. Its hard plastic casing and header pins must be decased (removing the plastic shell and direct-soldering wires to the PCB) to fit the strict mass and volume constraints of this weight class.

6. Mass Budget (300g Limit)

Maintaining the 300g strict limit requires careful material selection. The chassis is typically 3D printed from impact-resistant polymers (TPU for shock absorption or Nylon/Polycarbonate for rigidity). The components below must all fit within the weight constraint:

- Chassis & Armor (TPU/Nylon)
- Weapon Blade (Steel or Titanium)
- Weapon Motor (Drone BLDC)
- Battery (2S LiPo, ~300mAh)
- Drive Motors (2x N20 + Mounts)
- Wheels & Hubs
- FlySky FS-iA10B Receiver (Decased)
- Weapon ESC + Drive ESC
- Wiring, Switches, & Hardware

Total Estimated Mass: 270g / 300g Max

Note: The remaining 30g serves as an engineering tolerance for heavier wiring, structural reinforcements, or center-of-mass balancing weights.

